

Unit-1

Introduction to System Administration & IT Infrastructure Services

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Unit-1 Syllabus

- 1.1 Introduction to System Administration
- 1.2 Role and Responsibilities of a System Administrator
- 1.3 Importance of System Administration in IT Infrastructure
- 1.4 History and Evolution of System Administration
- 1.5 IT Infrastructure Components (Hardware and Software)
- 1.6 Types of IT Services (Web, Email, Database, etc.)
- 1.7 Service Level Agreements (SLAs)
- 1.8 IT Service Management (ITSM) Frameworks

System Administration

System Administration is the process of managing, maintaining, and configuring computer systems and servers to ensure they operate efficiently and securely.

Responsibilities:

- Installing and configuring operating systems (Windows, Linux)
- Managing users and permissions
- Monitoring system performance
- Performing backups and recovery
- Ensuring system security and updates

IT Infrastructure

IT Infrastructure refers to all the physical and virtual resources required to operate an organization's IT environment.

Components of IT Infrastructure:

- Hardware → Servers, computers, routers
- Software → Operating systems, applications
- Network → Internet, LAN, WAN
- Data Storage → Databases, cloud storage
- Security Systems → Firewalls, antivirus

IT Infrastructure Services

These are services that support and manage IT infrastructure.

Main Types of Services:

- Network Management → Managing connectivity and performance
- Cloud Services → Hosting applications and data online
- Storage Services → Data backup and recovery
- Security Services → Protecting systems from threats
- Support Services → Helpdesk and troubleshooting

System Administration

System Administration is the process of managing, maintaining, and supporting computer systems, servers, and networks to ensure they run efficiently, securely, and reliably.

Key Idea:

System Administration = Keeping IT systems running smoothly

Role of a System Administrator

A System Administrator (SysAdmin) is an IT professional responsible for managing, maintaining, and ensuring the smooth operation of computer systems, servers, and networks within an organization.

Key Idea:

System Administrator = Person who keeps computer systems running properly and securely

In a Company, a SysAdmin:

- Creates login accounts for employees
- Fixes network or system problems
- Keeps servers running 24/7
- Protects data from hackers

Roles of a System Administrator

1 System Manager

- Manages servers and computer systems
- Ensures systems are running properly

2 User Manager

- Creates and manages user accounts
- Controls access and permissions

3 Security Manager

- Protects systems from viruses and hackers
- Manages firewalls, antivirus, and updates

4 Network Manager

- Maintains network connections
- Ensures internet and communication systems work

Roles of a System Administrator (Continued)

5 Troubleshooter

- Identifies and fixes system problems
- Provides technical support to users

6 Backup Manager

- Takes regular data backups
- Restores data when needed

7 Performance Monitor

- Checks system speed and performance
- Optimizes system efficiency

8 Software Manager

- Installs and updates software
- Ensures applications work correctly

Responsibilities of System Administrator

1 System Installation & Configuration

- Install operating systems (Windows Server, Linux)
- Configure hardware (servers, storage devices)
- Set up system services and applications
- Perform initial system testing after setup
- **Goal:** Make systems ready for use

2 User & Access Management

- Create, modify, and delete user accounts
- Assign roles and permissions (access control)
- Manage authentication (passwords, policies, MFA)
- Maintain user groups and privileges
- **Goal:** Ensure correct access for the right users

Responsibilities of System Administrator (Continued)

3 System Monitoring & Performance Management

- Monitor CPU, memory, disk usage
- Track system logs and alerts
- Identify bottlenecks and performance issues
- Optimize system performance
- **Goal:** Keep systems fast and responsive

4 Security Management

- Install and update antivirus and firewall systems
- Apply security patches and updates
- Monitor for unauthorized access or threats
- Implement security policies and encryption
- **Goal:** Protect systems from cyber threats

Responsibilities of System Administrator (Continued)

5 Backup & Disaster Recovery

- Schedule regular data backups
- Test backup reliability
- Restore systems in case of failure
- Create disaster recovery plans
- **Goal:** Prevent data loss and ensure quick recovery

6 Network Management

- Configure and maintain network devices (routers, switches)
- Monitor network performance and connectivity
- Troubleshoot network issues
- Ensure secure communication
- **Goal:** Maintain reliable connectivity

Responsibilities of System Administrator (Continued)

7 Troubleshooting & Technical Support

- Diagnose hardware and software problems
- Resolve system crashes and errors
- Provide support to users (helpdesk role)
- Document issues and solutions
- **Goal:** Minimize downtime

8 Software & Patch Management

- Install and update software applications
- Apply system patches and updates
- Ensure compatibility and stability
- **Goal:** Keep systems up-to-date and secure

Responsibilities of System Administrator (Continued)

9 Documentation

- Maintain system configuration records
- Document procedures and policies
- Keep logs of changes and updates
- **Goal:** Ensure clarity and consistency

10 Automation & Scripting

- Write scripts (Shell, PowerShell)
- Automate repetitive tasks (backups, updates)
- Improve efficiency
- **Goal:** Save time and reduce human error

Responsibilities of System Administrator (Continued)

11 Cloud & Virtualization Management

- Manage virtual machines and cloud services
- Monitor cloud resources
- Optimize usage and cost
- **Goal:** Efficient use of modern infrastructure

12 Compliance & Policy Enforcement

- Ensure systems follow organizational policies
- Maintain compliance with regulations
- Conduct audits and security checks
- **Goal:** Meet legal and organizational standards

Importance of System Administration in IT Infrastructure

1 Ensures System Availability (Uptime)

- Keeps servers and systems running 24/7
- Minimizes downtime and service interruptions
- Quickly resolves failures
- **Why it matters:** Businesses depend on continuous system availability for operations and services

2 Enhances Security

- Protects systems from cyber threats (viruses, hackers, malware)
- Applies security patches and updates
- Manages firewalls, access control, and encryption
- **Why it matters:** Prevents data breaches and protects sensitive information

Importance of System Administration in IT Infrastructure

③ Improves Performance & Efficiency

- Monitors CPU, memory, and storage usage
- Optimizes system performance
- Removes bottlenecks
- **Why it matters:** Faster systems = higher productivity and better user experience

④ Supports Data Management

- Organizes and stores data properly
- Ensures regular backups
- Provides data recovery in case of failure
- **Why it matters:** Prevents data loss and ensures business continuity

Importance of System Administration in IT Infrastructure

5 Enables Scalability

- Adds new users, systems, and resources easily
- Supports business growth
- Adapts to changing needs
- **Why it matters:** Organizations can expand without system breakdowns

6 Provides Reliable Network Management

- Maintains network connectivity
- Ensures stable communication between systems
- Fixes network issues
- **Why it matters:** Smooth communication is essential for daily operations

Importance of System Administration in IT Infrastructure

7 Supports Business Continuity & Disaster Recovery

- Prepares disaster recovery plans
- Restores systems after failures or attacks
- Reduces downtime during emergencies
- **Why it matters:** Keeps business running even during unexpected events

8 Ensures Compliance & Policy Enforcement

- Maintains IT policies and standards
- Helps meet legal and regulatory requirements
- Conducts audits and monitoring
- **Why it matters:** Avoids legal issues and ensures proper system use

Importance of System Administration in IT Infrastructure

9 Reduces Operational Costs

- Prevents major system failures
- Optimizes resource usage
- Automates repetitive tasks
- **Why it matters:** Saves time and money for the organization

10 Supports Users and Business Operations

- Provides technical support to employees
- Ensures systems are user-friendly and accessible
- Solves issues quickly
- **Why it matters:** Improves productivity and employee satisfaction

Importance of System Administration in IT Infrastructure

11 Enables Modern Technologies

- Supports cloud computing and virtualization
- Manages advanced IT infrastructure
- Integrates new technologies
- **Why it matters:** Keeps organizations competitive in the digital world

Example (In a Company):

- System administrators keep servers running
- Protect company data from cyber attacks
- Ensure employees can access systems anytime
- Recover data if systems crash

History and Evolution of System Administration

1. Early Era (1950s–1970s): Birth of System Administration

- Computers were mainframes used by governments and universities
- System administration was done by operators and engineers (not a separate role)
- Tasks included:
 - Loading programs using punch cards
 - Monitoring hardware
 - Managing batch jobs
- No real “sysadmin” concept yet — only machine caretakers

History and Evolution of System Administration

2. Time-Sharing & Minicomputers (1970s–1980s)

- UNIX systems emerged
- Multiple users could access one system simultaneously
- Introduction of:
 - User accounts
 - File permissions
 - Basic security mechanisms
- The system administrator role begins to form during this era

History and Evolution of System Administration

3. Personal Computer Era (1980s–1990s)

- Rise of PCs like MS-DOS and later Microsoft Windows
- Organizations started building local networks (LANs)
- Admin responsibilities expanded:
 - Installing software
 - Managing printers and networks
 - Basic troubleshooting
- Sysadmins became essential in businesses

History and Evolution of System Administration

4. Internet Boom (1990s–2000s)

- Growth of the internet and web servers
- Popular systems like Linux gained importance
- New responsibilities:
 - Web server management (Apache)
 - Email servers
 - Network security (firewalls)
- Security and uptime became critical concerns

History and Evolution of System Administration

5. Enterprise & Virtualization Era (2000s–2010s)

- Large-scale IT infrastructures emerged
- Introduction of virtualization tools like VMware
- Admin tasks expanded:
 - Managing virtual machines
 - Storage systems
 - Backup and disaster recovery
- Efficiency improved — one machine could run many systems

History and Evolution of System Administration

6. Cloud Computing Era (2010s–Present)

- Shift from physical servers to cloud platforms:
 - Amazon Web Services (AWS)
 - Microsoft Azure
- Infrastructure became on-demand
- Admin role evolved into:
 - Cloud engineers
 - DevOps engineers
- Focus shifted from hardware to services and scalability

History and Evolution of System Administration

7. Automation & DevOps Era (Modern)

- Rise of DevOps culture (Development + Operations)
- Tools like:
 - Docker (containers)
 - Kubernetes
- Key trends:
 - Infrastructure as Code (IaC)
 - Continuous Integration/Deployment (CI/CD)
 - Automated monitoring
- Sysadmins now write scripts and automate everything

8. Future Trends (AI & Self-Healing Systems)

- Use of AI for:
 - Predictive maintenance
 - Auto-scaling systems
- “Self-healing” infrastructure can fix issues automatically
- System administration is becoming more intelligent and proactive

IT Infrastructure Components

IT Infrastructure is the complete set of physical and virtual resources that support computing, storage, networking, and services in an organization. It is broadly divided into:

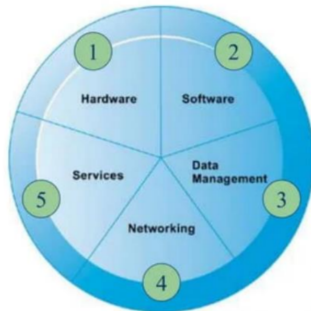
- **Hardware (Physical Layer)**
- **Software (Logical Layer)**

Both layers work together to deliver IT services.

IT Infrastructure Components

IT Infrastructure Components

- An IT infrastructure is composed of hardware, software, networking technology, data management technology, and technology services.



IT Infrastructure Components (Hardware & Software)

IT Infrastructure is the complete set of physical and virtual resources that support computing, storage, networking, and services in an organization.

It is broadly divided into:

- **Hardware (Physical Layer)**
- **Software (Logical Layer)**

Both layers work together to deliver IT services.

Hardware Components (Physical Infrastructure)

Hardware includes all tangible equipment used to process, store, and transmit data.

1. Servers

Servers are high-performance computers that provide services to clients.

Types:

- File Server
- Web Server
- Database Server
- Application Server

Features:

- High CPU power
- Large RAM
- Redundant power supply

2. Client Devices

Devices used by end-users to access IT services.

Examples:

- Desktop computers
- Laptops
- Smartphones
- Tablets

Functions:

- Send requests to servers
- Display output to users

Hardware Components (Physical Infrastructure)

3. Networking Devices

These devices enable communication between systems.

Devices:

- Router → Connects networks
- Switch → Connects LAN devices
- Firewall → Protects network
- Access Point → Wireless connectivity

Functions:

- Data transmission
- Traffic management
- Network security

4. Storage Devices

Used to store data permanently or temporarily.

Types of Storage:

- Primary → RAM
- Secondary → HDD, SSD
- Network Storage → NAS, SAN

Advanced Storage:

- Cloud storage
- Backup storage systems

Hardware Components (Physical Infrastructure)

5. Data Centers

A data center is a facility that houses IT infrastructure.

Components Inside Data Center:

- Servers
- Storage systems
- Networking equipment
- Cooling systems
- Power backup (UPS, generators)

Purpose:

- Ensures high availability
- Provides reliability

6. Input/Output Devices

Input Devices:

- Keyboard
- Mouse
- Scanner

Output Devices:

- Monitor
- Printer

Software Components (Logical Infrastructure)

Software provides instructions that tell hardware what to do and how to do it.

1. Operating Systems (OS)

The OS is the core software that manages hardware and software resources.

Functions:

- Process management
- Memory management
- File system management
- Device control

Examples:

- Windows Server
- Linux (Ubuntu, Red Hat)
- macOS

2. Application Software

Programs used by users to perform tasks.

Examples:

- Office tools (Word, Excel)
- Web browsers (Chrome)
- Business applications (ERP, CRM)

Software Components (Logical Infrastructure)

3. System Software

Supports the operation of the computer system.

Includes:

- Device drivers
- Utility programs
 - Disk cleanup tools
 - Backup tools

4. Database Management Systems (DBMS)

Used to store, manage, and retrieve structured data.

Functions:

- Data storage
- Query processing
- Data security

Examples:

- MySQL
- Oracle Database
- SQL Server

Software Components (Logical Infrastructure)

5. Middleware

Acts as a bridge between different applications or systems.

Functions:

- Enables communication between software
- Integrates different platforms

Example:

- Connecting a web application to a database

6. Security Software

Protects systems and data from threats.

Types:

- Antivirus software
- Firewall (software-based)
- Encryption tools
- Intrusion Detection Systems (IDS)

Software Components (Logical Infrastructure)

7. Virtualization Software

Allows multiple virtual systems to run on one physical machine.

Examples:

- VMware
- VirtualBox

Benefit:

- Improves resource utilization
- Increases flexibility

8. Cloud Platforms

Provide IT services over the internet.

Examples:

- AWS
- Microsoft Azure
- Google Cloud

Uses:

- Storage
- Computing
- Hosting

Types of IT Services (Web, Email, Database, etc.)

IT services are the core services provided by IT infrastructure that allow users and organizations to communicate, store data, and run applications efficiently.

Types of IT Services

1. Web Services

Web services allow users to access information and applications through the internet or intranet using web browsers.

Functions:

- Hosting websites and web applications
- Delivering content over the internet
- Supporting online services (e-commerce, portals)

Examples:

- Websites (Google, Amazon)
- Online banking systems
- Learning management systems (LMS)

Types of IT Services

2. Email Services

Email services enable users to send, receive, and manage electronic messages.

Functions:

- Sending and receiving emails
- Attachment sharing
- Spam filtering and security

Examples:

- Gmail
- Outlook
- Yahoo Mail

3. Database Services

Database services manage and store structured data for applications and users.

Functions:

- Data storage and retrieval
- Data security and backup
- Query processing

Examples:

- MySQL databases
- Oracle Database
- Microsoft SQL Server

Types of IT Services

4. File Storage Services

These services allow users to store and share files securely.

Functions:

- Data storage and sharing
- Backup and synchronization
- Remote access to files

Examples:

- Google Drive
- Dropbox
- OneDrive

5. Cloud Computing Services

Cloud services provide computing resources over the internet.

Types:

- IaaS (Infrastructure as a Service)
→ Virtual machines, storage
- PaaS (Platform as a Service) →
Development platforms
- SaaS (Software as a Service) →
Applications like Gmail, Office 365

Examples:

- AWS
- Microsoft Azure
- Google Cloud

Types of IT Services

6. Network Services

Network services manage communication between devices and systems.

Key Functions:

- IP address management
- Domain name resolution
- Network connectivity

Examples:

- DNS (Domain Name System)
- DHCP (Dynamic Host Configuration Protocol)
- VPN services

7. Security Services

Security services protect IT systems and data from threats.

Functions:

- Virus and malware protection
- Firewalls and intrusion detection
- Data encryption

Examples:

- Antivirus systems
- Firewalls
- Security monitoring tools

Types of IT Services

8. Application Services

Application services support business applications used by organizations.

Key Functions:

- Running enterprise applications
- Supporting business processes
- Integrating different systems

Examples:

- ERP systems
- CRM systems
- HR management systems

9. Backup and Recovery Services

These services ensure data is safe and can be restored after failure.

Key Functions:

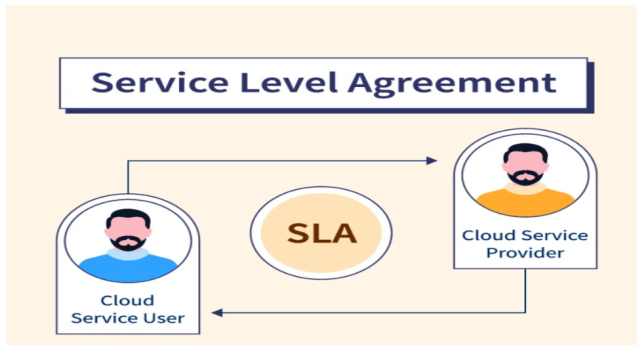
- Data backup scheduling
- Disaster recovery
- System restoration

Examples:

- Cloud backup services
- Tape backup systems

SLA





The Goals of Service Level Agreement (SLA)



The Goals of Service Level Agreement



Importance of SLAs (Service Level Agreements)

- Ensures reliable IT service delivery
- Improves customer satisfaction
- Builds trust between provider and client
- Helps in performance monitoring
- Reduces service disputes

Real-Life Example: Internet Service Provider (ISP)

- Promises 99.9% uptime
- Guarantees support within 2 hours
- Provides compensation if downtime exceeds limit

Components of SLA (Service Level Agreement)

● 1. Service Description

- Defines what services are provided
- Example: email service, web hosting, database support

● 2. Performance Metrics

- Measures service quality and performance
- Uptime (e.g., 99.9% availability)
- Response time (how fast issues are handled)
- Resolution time (how quickly problems are fixed)

● 3. Responsibilities

- **Service Provider:** Maintenance, support, ensuring service quality
- **Customer:** Proper usage and reporting issues

● 4. Availability

- Specifies when services are available
- Example: 24/7 system access or business hours support

Components of SLA (Service Level Agreement)

- **5. Support & Maintenance**

- Helpdesk support hours
- System updates and maintenance schedules

- **6. Security Requirements**

- Data protection rules
- Backup policies
- Access control standards

- **7. Penalties and Compensation**

- Defines consequences if service levels are not met
- Example: discounts, penalties, or service credits

Types of SLAs (Service Level Agreements)

- **1. Customer-based SLA**

- Agreement with a specific customer
- Covers all services provided to that customer

- **2. Service-based SLA**

- Same service offered to all customers
- Example: cloud storage SLA for all users

- **3. Internal SLA**

- Between departments within the same organization
- Example: IT department supporting HR department

IT Service Management (ITSM) Frameworks

IT Service Management (ITSM) is a structured approach to designing, delivering, managing, and improving the way IT services are used within an organization.

Instead of focusing just on technology, ITSM focuses on delivering value to users and the business.

ITSM = Delivering IT as a reliable service (not just managing technology)

Objectives of ITSM

- Align IT services with business goals
- Improve service quality and reliability
- Enhance customer/user satisfaction
- Reduce costs and downtime
- Ensure continuous improvement

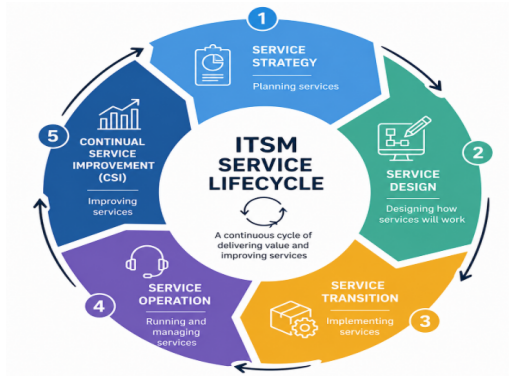
IT Service Management (ITSM) Frameworks

Service

A service is a way of delivering value to users.

Example: Email service, cloud storage, website hosting

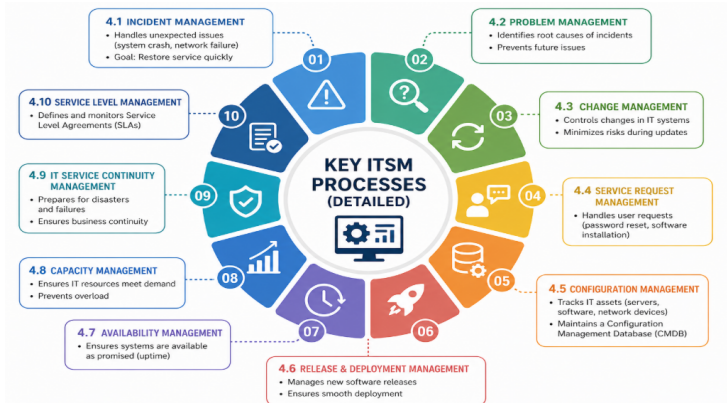
Service Lifecycle



Components of ITSM

- **1. Service Strategy**
 - Defines what services to offer
 - Aligns services with business goals
- **2. Service Design**
 - Plans how services will work
 - Includes architecture, processes, and security
- **3. Service Transition**
 - Ensures smooth rollout of new or changed services
 - Includes testing and deployment
- **4. Service Operation**
 - Daily IT service activities
 - Help desk support, incident resolution, system monitoring
- **5. Continual Service Improvement (CSI)**
 - Ongoing optimization of services
 - Based on performance data and user feedback

IT Service Management (ITSM) Frameworks



- **ITIL (Information Technology Infrastructure Library)**

- Most widely used ITSM framework
- Focuses on best practices for IT service delivery and management

- **COBIT**

- Governance-focused framework
- Helps align IT with business goals and risk management

- **ISO/IEC 20000**

- International standard for IT Service Management
- Defines requirements for establishing and improving ITSM systems

COBIT (Control Objectives for Information and Related Technologies)

COBIT is a framework developed by ISACA for IT governance and management. It helps organizations ensure that Information Technology (IT) supports business goals, manages risks, and uses resources effectively.

COBIT is a globally recognized framework used to:

- Govern and manage enterprise IT
- Align IT with business objectives
- Improve security and compliance
- Manage IT risks
- Measure IT performance

Main Purpose of COBIT

The purpose of COBIT is to:

- Align IT with business goals
- Deliver value from IT investments
- Manage IT-related risks
- Ensure security and compliance
- Improve IT performance and quality
- Optimize resource utilization

COBIT: Governance vs Management

COBIT clearly separates two concepts:

Governance (Top Level – Board/Executives)

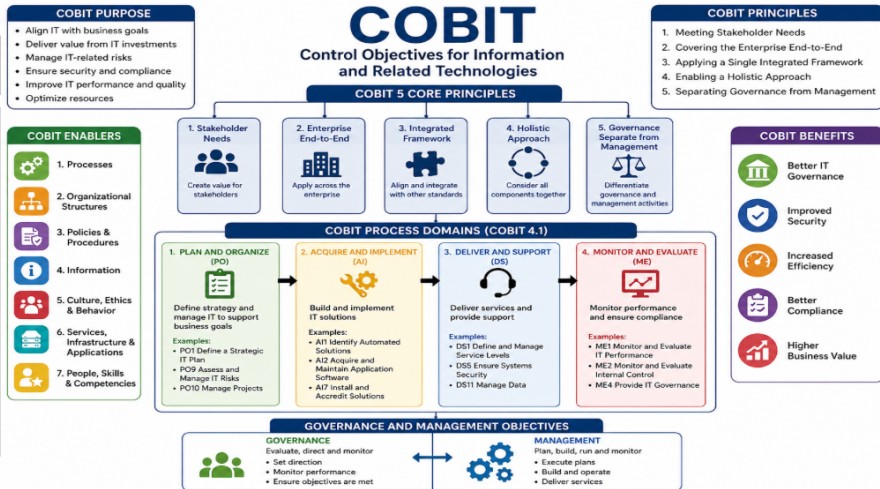
- Sets direction
- Evaluates performance
- Ensures organizational goals are achieved

Management (Operational Level – IT Team)

- Plans and executes work
- Implements processes
- Delivers IT services

COBIT connects governance and management smoothly.

COBIT (Control Objectives for Information and Related Technologies)



Governance & Management Objectives in COBIT

COBIT defines 40 objectives grouped into 5 domains:

1. EDM (Evaluate, Direct, Monitor) – Governance Domain

- Ensures IT delivers value
- Example:
 - Evaluate IT strategy
 - Monitor performance

1. APO (Align, Plan, Organize) – Management Domain

- Focuses on strategy and planning
- Example:
 - IT budgeting
 - Risk management

COBIT Governance & Management Domains (Continued)

2. BAI (Build, Acquire, Implement)

- Focuses on development and implementation
- Example:
 - Software development
 - System upgrades

3. DSS (Deliver, Service, Support)

- Covers daily IT operations
- Example:
 - Help desk support
 - Incident handling

4. MEA (Monitor, Evaluate, Assess)

- Focuses on performance tracking and compliance
- Example:
 - Audits
 - KPI measurement

COBIT (Control Objectives for Information and Related Technologies)

COBIT CAPABILITY LEVELS (0-5) Measuring Process Maturity from Incomplete to Optimizing						
	LEVEL 0 INCOMPLETE PROCESS	LEVEL 1 PERFORMED PROCESS	LEVEL 2 MANAGED PROCESS	LEVEL 3 ESTABLISHED PROCESS	LEVEL 4 PREDICTABLE PROCESS	LEVEL 5 OPTIMIZING PROCESS
DESCRIPTION	0 Process is not implemented at all or fails to achieve its purpose 	1 Process exists and works, but: Informal Unstructured Success depends on individuals 	2 Process is: Planned Monitored Documented (partially) 	3 Process is: Standardized Well documented Used across organization 	4 Process is: Measured Controlled using data (KPIs) 	5 Process is: Continuously improved Innovative (automation, AI, tools) 
EXAMPLE	No formal system for handling IT incidents Employees fix issues randomly	IT team resolves issues, but no documentation or standard steps	Incident handling process exists Tasks are assigned and tracked	Standard IT support procedures used by all teams	SLA: "Resolve incidents within 2 hours" Performance is tracked and predictable	Automated incident detection and resolution Continuous feedback loop
OUTPUT	 No reliable results	 Basic results, but inconsistent	 More consistent and controlled	 Reliable and repeatable results	 Stable, measurable performance	 Maximum efficiency and improvement
KEY CHARACTERISTICS	- No process - No structure - No consistency - Ad-hoc - Reactive	- Informal - Unstructured - Depends on individuals - Reactive	- Planned - Monitored - Partially documented - Basic control	- Standardized - Well documented - Consistent execution - Proactive	- Measured - Data-driven - Controlled - Predictable - Proactive	- Continuously improved - Innovative - Automated - Optimized - Proactive
 MATURITY GROWTH  COBIT Capability Levels help organizations assess and improve their processes for better governance, control and value delivery.						

ISO/IEC 20000 Introduction

ISO/IEC 20000 is the international standard for IT Service Management (ITSM).

It defines requirements for organizations to plan, establish, implement, operate, monitor, review, maintain, and improve IT services.

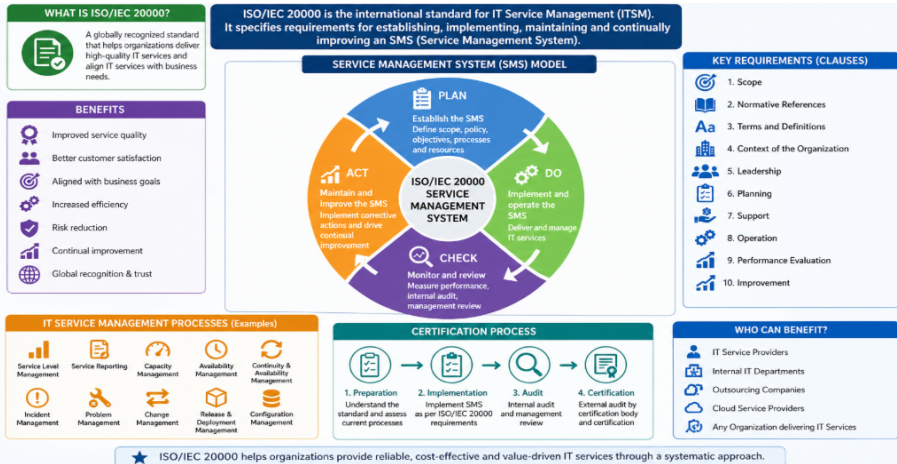
It ensures:

- High-quality IT services
- Standardized processes
- Continuous improvement
- Global certification

Objectives of ISO/IEC 20000

- Deliver consistent and reliable IT services
- Align IT services with business needs
- Improve customer satisfaction
- Ensure effective risk management
- Establish a continual improvement culture

COBIT (Control Objectives for Information and Related Technologies)



ITIL (Information Technology Infrastructure Library)

ITIL (Information Technology Infrastructure Library) is a globally recognized framework of best practices for IT Service Management (ITSM). It provides structured guidelines on how to design, deliver, manage, and improve IT services so they align with business needs.

In simple terms, ITIL helps organizations ensure that IT services (like networks, servers, software, and help desks) run efficiently and reliably.

Purpose of ITIL:

- Improve the quality of IT services
- Reduce downtime and service failures
- Increase customer/user satisfaction
- Standardize IT processes
- Align IT services with business goals

ITIL Processes

ITIL Processes (Examples):

- Incident Management → Fixing issues quickly (e.g., internet not working)
- Problem Management → Finding root cause of repeated issues
- Change Management → Handling system updates safely
- Service Desk → Single point of contact for users
- Configuration Management → Tracking IT assets and systems

Benefits of ITIL:

- Better service quality and reliability
- Faster problem resolution
- Reduced IT costs
- Improved communication between IT teams
- More efficient use of resources

ITIL Example in Real Life

Example: University Wi-Fi Issue

- Incident is reported to service desk
- IT team resolves it quickly (Incident Management)
- Root cause is analyzed (Problem Management)
- Future improvements are planned (CSI)